

SECTION B

7 (**short**) A satellite is in an elliptical orbit around the earth. The eccentricity of the orbit is 0.7 and the radius of the earth is R . The altitude of the satellite above the earth's surface at the apogee is $11R$ and the speed of the satellite at the apogee is V .

- (a) Find the altitude at the perigee. [5]
- (b) Find the speed of the satellite at the perigee. [5]

8 (**short**)

(a) A ball of mass M and initial speed u collides, head-on, with a stationary ball of mass m . Assuming the collision is perfectly elastic, find the velocity, V , of the second ball after the collision. [6]

(b) A third ball of mass m_3 is placed, initially at rest, between the two original balls. The collision now proceeds in two steps: first M strikes m_3 , then m_3 strikes m . Again assuming the collisions are head-on and elastic, find the value of m_3 which results in the largest final speed for m . [4]

[Hint: Maximizing a quantity, y , is equivalent to minimizing its reciprocal, $1/y$.]